

multiple A number is described as “a multiple of x” when x is multiplied by 2, 3, 4, or any other whole number.

mutation A random change in a chromosome of a cell, either to a particular gene or on a larger scale.

myelin Fatty material wrapped around some neurons, which speeds up the transmission of signals.

myofibril Any of the tiny structures in a muscle that enable it to contract.

nano- A prefix meaning a billionth (thousand-millionth).

nanometer A billionth of a meter.

natural selection The process whereby the inheritable characteristics that increase one’s chances of survival and reproduction are passed on to the next generation.

Neanderthal A member of an extinct species closely related to modern humans.

nebula Originally, a term for any distant cloudlike object visible beyond Earth’s atmosphere. It now applies specifically to huge clouds of dust and gas which are often the locations for new stars to be formed.

negative number A number less than zero.

nephron One of the million or so purification and filtration units in the kidney.

nerve A cablelike structure transmitting information and control instructions in the body. A typical nerve consists of strands of many separate nerve cells (*neurons*) running parallel to, but insulated from each other.

nervous system The network of nerve cells (including the brain) that controls the body.

neuron A nerve cell.

neutrino A tiny, almost massless, uncharged subatomic particle abundant in the Universe but rarely interacting with other matter.

neutron A subatomic particle found in all atomic nuclei except the normal form of hydrogen. It is similar in size to a proton but has no electric charge.

neutron star A small but extremely dense star made mostly of neutrons, formed by the gravitational collapse of a giant star.

newton The SI unit of force.

nitrogen A chemical element that as a gas makes up most of Earth’s atmosphere, and combined into compounds is essential to living things.

noble gases Gases such as helium and neon that have a complete complement of electrons in their outer shell and are very unreactive.

nomenclature A body or system of names.

nuclear fission See *fission*.

nuclear fusion A reaction in which the nuclei of light atoms such as hydrogen fuse to form a heavier nucleus, releasing energy.

nuclear reaction A change in the nucleus of an atom.

nucleolus A small, dense, round body inside the nucleus of a cell.

nucleus (1) Central part of an atom, made up of protons and neutrons. (2) The structure in eukaryote cells that contains *chromosomes*.

nutrients Substances that are used by living organisms for growth, maintenance, and reproduction.

observatory A building or institution where astronomers study space.

ohm The SI unit of electrical resistance.

Oort cloud Huge spherical region containing comets that is thought to exist toward the outer boundary of the Solar System.

opiates Drugs related to opium. They are strong painkillers but have many side-effects.

optical fibers Thin glass fibers along which light travels, used in communication.

optics The study of the behavior of light and how it is affected by devices such as lenses, mirrors, etc.

orbit The path of any body that is circling around another one.

orbital period The time a celestial body takes to complete one *orbit* around another.

organ A group of tissues, usually grouped together in a discrete structure, that has a special function, such as the brain.

organic Adjective that can refer to (1) Any compound containing carbon, with the exception of some simple molecules such as carbon dioxide, (2) Food produced without the use of artificial fertilizers or pesticides.

Orrery A mechanical model of the Solar System, showing the relative positions and orbits of the planets and their moons.

oscillation A regular movement back and forth.

oscillator A circuit or instrument that produces an alternating current of known frequency.

oscilloscope An instrument that shows electrical signals on a screen.

osmosis The movement of water through a semipermeable membrane from a weak solution to a more concentrated one.

ovary (1) The structure in animals that produces female sex cells (gametes). (2) A specialized region of the female part of a flower that contains the ovules.

ovule A structure within a flower that develops into a seed after fertilization.

ovum An egg cell.

oxidation Originally, a reaction where a substance combines with oxygen; now used for any reaction where a substance loses electrons. Its opposite is reduction.

oxide A compound of oxygen with one other element.

oxidizing agent A compound that can cause oxidation.

oxygen A reactive gas that makes up 21 percent of Earth’s atmosphere and is essential to life.

ozone A very reactive form of oxygen that has three atoms in each *molecule* instead of two.

P wave Primary wave. A fast-moving earthquake wave that alternately stretches and squeezes rocks as it moves.

palynology The study of living and fossil pollen grains and spores.

pancreas A gland close to the stomach that secretes digestive enzymes and also hormones that regulate glucose levels.

parallax The apparent movement of objects against each other when an observer moves

position, such as nearby trees against distant hills. Astronomers use the same principle to measure the distance to nearby stars.

parallel circuit A circuit in which there are at least two independent paths to get back to the source.

particle In physics, usually short for subatomic particle.

particle accelerator A giant machine in which subatomic particles are accelerated along or around a tunnel by electromagnets, and smashed together at very high speeds.

particle physics The branch of physics that deals with subatomic particles.

pasteurization The heating of food to destroy disease-causing bacteria.

pathogen A disease-causing microorganism.

peptide A *molecule* similar to a protein in structure but typically smaller.

pericardium A tough double-layered membrane surrounding the heart.

periodic table A table of the chemical elements in order of their atomic numbers, whose vertical columns bring together elements with similar properties.

perpetuum mobile Also called perpetual motion, the theoretically impossible notion that a machine can be made to run forever without energy input although work is extracted from it.

petal One of a set of structures surrounding the sexual organs of most flowers, usually shaped and colored for attracting pollinating animals.

pH A measure of acidity or alkalinity of a solution. A pH of 7 is neutral, below 7 is *acid*, and above 7 is alkaline.

pharmacology The study of drugs and how they act in the body.

phlogiston theory A now-disproved 18th-century theory that all burning involved giving off a substance called “phlogiston.”

photoelectric effect The emission of electrons from the surfaces of some objects when light hits them.

photon The particle that makes up light and other electromagnetic radiation.

photoperiodism Any situation where life-processes in living things are affected by the length of daylight they are exposed to.

photosynthesis The process by which plants and algae make food from water and carbon dioxide, using energy from the Sun.

physiology The study of body processes. Also a term for the body processes.

pi The ratio of the circumference of a circle to its diameter, approximately 22 divided by 7, or about 3.14159.

piezoelectric effect The production of electricity by applying mechanical stress to certain crystals, such as quartz.

pistil A female organ of a flower.

piston A close-fitting sliding disk or short solid cylinder attached to a rod that is pushed up and down in the cylinder of an engine to provide power.

pitch (1) The property of a sound that makes it high or low. (2) The angle of an aeroplane wing, propeller blade, etc.

Planck constant Symbol *h*, the ratio of the energy in one photon of electromagnetic radiation to its frequency. It is a fundamental constant in quantum physics.

plane figure A two-dimensional shape.

planet A large spherical or almost spherical body that orbits a star. See also *dwarf planet*.

plankton Plants, animals, or other life forms living in open water that cannot swim strongly and so drift with the currents. Most are small or microscopic.

plant A member of one of the major kingdoms of living things, making their own food using photosynthesis. They include everything from trees and flowers to ferns and mosses, but not most algae (see *alga*).

plasmid A normally circular strand of DNA in bacteria or protozoa.

plate tectonics The theory and phenomena of Earth’s *lithosphere* being divided into huge rigid plates that move with respect to each other. Some plates include continents or parts of continents, while others comprise only deep ocean floor.

platelets Irregular disk-shaped microscopic structures in the blood which function to coagulate the blood and stop bleeding.

pluripotent Term applied to a stem cell that can give rise to any of several other cell types.

pneumatics Branch of physics studying the mechanical properties of air and other gases.

polarized light Light in which the wave vibrations occur only in one plane.

pollen tube The tube that a germinating pollen grain forms as it grows down the female part of a flower to fertilize an ovule.

pollination The deposition of pollen on a flower so that its ovules can be fertilized and it can set seed.

polymer A long, thin *molecule* made of many identical or very similar small molecules joined together; also, a substance made of such molecules.

population In biology, a group of individuals of the same species, especially when able to interbreed with each other.

positron Positively charged counterpart of an electron, sometimes called antielectron.

potential difference The electrical equivalent of pressure. A high potential difference is like a high pressure forcing electricity round a *circuit*. Also called voltage.

potential energy Stored energy that a body has because of its position or internal state.

power (1) The rate of change of energy. (2) The number of times a number is multiplied by itself: for example, $x \times x \times x$, or x^3 , is also called “x to the power of 3”.

precipitate Tiny solid particles formed in a liquid as a result of a chemical reaction.

predator An animal that feeds by attacking and eating other animals (its prey), especially those that are relatively large in relation to its own size.

preservation The process of keeping something in its original state, or free from harm, erosion, or decay.

pressure Continual physical force pushing against an object, especially considered as force per unit area.